

PMD · Project Roadmap Architecture

Artifact #18 · v2.0 · May 2026 · 8 Phases · 66+ Projects · PMD System · Internal

ARCHITECTURE INTENT

PORTFOLIO MILESTONE MAP

Project	Type	Phase	Why It Works
P5	Product Teardown & Redesign Case Study	I	PM thinking visible before a single backend line exists. Low technical dependency, high product thinking signal.
P33	Cross-Functional GTM — Pricing Launch	V	Pricing decisions require engineering, marketing, sales, and legal alignment. Full stakeholder coordination artifact.
P36	Metric Optimization — Activation Funnel	V	Before/after metrics, funnel analysis, and experiment design — the clearest PM impact story.
P43	GTM — Integrations Marketplace Launch	VI	Platform and ecosystem strategy thinking. Integration design + OAuth2 + GTM kit.
P50	0-to-1 Greenfield — AI PRD Assistant	VII	Solves a problem hiring managers recognize. Full 0-to-1 cycle with AI integration, PRD, and launch plan.
P64	Analytics PM — Executive Dashboard	VIII	Business intelligence, cohort analysis, and KPI storytelling at executive depth.
P65	Cross-Functional GTM — Pricing Migration	VIII	Enterprise GTM kit, stakeholder alignment artifacts, risk log, and pricing migration execution.
P66	Flagship Portfolio — Full SaaS Capstone	VIII	Cumulative synthesis of all prior capabilities. Production-grade SaaS with complete case study suite.

PHASE OVERVIEW

Phase	Title	Range	Projects	Core Capability
I	FOUNDATION	P1 – P7	7	Static Web, Git & Product Communication
II	FRONT-END FRAMEWORK	P8 – P16	9	React, Component Architecture & CMS
III	BACKEND & DATABASE	P17 – P24	8	Node, Custom APIs & Persistence
IV	PRODUCT COLLABORATION & TEAM FEATURES	P25 – P30	6	PRODUCT COLLABORATION & TEAM FEATURES
V	PAYMENTS, MONETIZATION & GROWTH ANALYTICS	P31 – P40	10	PAYMENTS, MONETIZATION & GROWTH ANALYTICS
VI	REAL-TIME, SCALE & ADVANCED INFRASTRUCTURE	P41 – P47	7	REAL-TIME, SCALE & ADVANCED INFRASTRUCTURE
VII	AI & MACHINE LEARNING INTEGRATION	P48 – P56	9	AI & MACHINE LEARNING INTEGRATION
VIII	PLATFORM, MOBILE & PRODUCTION SCALE	P57 – P66+	10	PLATFORM, MOBILE & PRODUCTION SCALE

Build the bedrock. Master HTML, CSS, JavaScript, Git, and deployment before touching any framework or library. Establish the product communication habit from day one: every project ships with a PRD, a problem statement, and at least one PM artifact. Introduce Figma for wireframing early. PM skills introduced: problem framing, persona writing, basic PRD, case study structure, value proposition design.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Personal Link-in-Bio / Profile Page				
P1	Single-page static site: profile section, curated links, social icons, and a goal statement. Fully responsive. First GitHub commit and Netlify deploy.	Problem framing · Single-user persona · Definition of done · First PRD	HTML5 semantic structure · CSS Flexbox · CSS custom properties · Git init · Netlify deploy pipeline	HTML5 · CSS3 · Git · GitHub · Netlify	PRD v1 · Persona · Deployed URL
	■ <i>First ship. Prove the CI/CD pipeline works end to end. Nothing else matters yet.</i>				
	Multi-Page Product Marketing Site				
P2	Multi-page marketing site for a fictional SaaS product: homepage, features, pricing, FAQ, and contact. Navigation, reusable layouts, and product messaging copy.	Information architecture · Value proposition design · Pricing page structure · Navigation UX	CSS Grid · Multi-page site structure · Tailwind CSS introduction · Vanilla JS modals and nav	HTML5 · CSS3 · Tailwind CSS · Vanilla JS · GitHub · Netlify	IA diagram · PRD · Wireframe (Figma intro)
	Interactive Product Calculator / ROI Estimator				
P3	Browser-based calculator — ROI estimator, pricing calculator, or productivity impact tool. User inputs, conditional logic, and results display. No backend.	Outcome-based product design · Input validation spec · Business logic documentation · User feedback states	DOM manipulation · Forms and input validation · Client-side conditional logic · Basic error handling	Vanilla JS · HTML5 · CSS3 · GitHub	Business logic spec · PRD · Edge case log
	■ <i>Products communicate value through measurable outcomes. This project internalizes that.</i>				
	Responsive Case Study Template				
P4	Reusable HTML/CSS case study page template with structured sections: problem, research, solution, metrics, tradeoffs, learnings. The format used for every future portfolio project.	Case study structure (5-part) · Content hierarchy · PM artifact presentation · Visual storytelling	Responsive layout · Print-friendly CSS · Semantic sectioning · Typography system	HTML5 · CSS3 · Figma (layout mockup) · GitHub · Netlify	Case study template · Figma mockup · PRD
	Product Teardown & Redesign Case Study				
	★ PORTFOLIO CANDIDATE — Product Teardown & Redesign				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P5	Select a widely-used app with visible UX friction. Conduct a heuristic evaluation, document the friction, design a data-backed redesign in Figma, and build a static HTML prototype of the improved flow.	UX critique methodology · Heuristic evaluation · Competitive analysis · Problem validation · OKR definition for design changes · Success metric design	Figma high-fidelity wireframes · Component-driven design · Static HTML prototype · axe-core accessibility audit	Figma · HTML5 · CSS3 · axe-core · GitHub	UX audit report · Figma mockup · Redesign PRD · OKRs · Case study
	■ <i>First portfolio milestone candidate. PM thinking is visible before a single line of backend code exists.</i>				
	Public API Data Dashboard				
P6	Dashboard consuming a public REST API (weather, finance, GitHub, movies). Multiple chart types, loading states, error states, and empty states handled explicitly.	API dependency risk documentation · Data visualization principles · Dashboard UX hierarchy · Third-party SLA awareness	Fetch API · Async/Await · REST API consumption · JSON parsing · Chart.js · Error and loading state patterns	Vanilla JS · Fetch API · Chart.js · OpenWeatherMap or CoinGecko API · GitHub	API integration spec · Error state matrix · Dashboard wireframe · PRD
	Multi-Page Portfolio Site with Vanilla JS Routing				
P7	Rebuild thepmdev.com foundation as a content-rich multi-page site with JS-driven navigation, animated transitions, dynamic content injection, and Open Graph tags.	IA design · Navigation strategy · SEO fundamentals · Content hierarchy · Personal brand positioning	History API / hash routing · JS module pattern · CSS transitions · Meta and OG tags · GA4 integration	Vanilla JS modules · History API · CSS transitions · Google Analytics 4 · GitHub · Netlify	IA diagram · Navigation spec · SEO checklist · PRD
	■ <i>Closes Phase I. Before touching React, understand what a framework is actually solving.</i>				

Graduate from document-oriented thinking to component-oriented thinking. React is the dominant front-end framework in product companies — master it deeply before adding a backend. Introduce headless CMS, managed auth, and cloud-hosted databases as a stepping stone to custom backends. PM skills introduced: design systems, component spec writing, content modeling, A/B hypothesis framing, migration planning.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	React Component Library & Design System				
P8	Reusable component library: Button, Input, Card, Modal, Toast, Badge, Dropdown. Document each component with props, states, and usage examples. Pure UI primitives — no app logic.	Component spec writing · Design token documentation · Reusability criteria · Accessibility requirements per component	React functional components · Props and PropTypes · CSS Modules · TypeScript introduction · Vite setup	React · Vite · CSS Modules · TypeScript · axe-core	Component spec · Design token inventory · Accessibility checklist · PRD
	React Task Board (Kanban) with State Management				
P9	Trello-style Kanban board: create, edit, delete, move tasks between columns, tag filtering, priority levels, due dates. State via useState and useReducer. LocalStorage persistence.	CRUD interaction design · Prioritization logic · Data model definition · Workflow design	useState · useReducer · useEffect · Drag-and-drop API · Conditional rendering · Component composition	React · Vite · TypeScript · Tailwind CSS · LocalStorage	Data model spec · Interaction spec · PRD
	■ Kanban is the most PM-relevant interface pattern. Build it, understand it deeply.				
	Product Requirements Manager App				
P10	Web app for creating lightweight PRDs: problem, users, requirements, success metrics, risks, open questions. Structured forms, Markdown editor, template system, LocalStorage persistence.	PM artifact design · Structured data modeling · Reusable template patterns · Information hierarchy	React controlled forms · Markdown editor integration · Template pattern · TypeScript interfaces	React · TypeScript · Tailwind CSS · Marked.js or similar · LocalStorage	PRD template spec · Data model · PRD
	■ A PM building a PRD tool is a PM who understands the artifact from the inside.				
	Now / Next / Later Roadmap Builder				
P11	Roadmap planning app: create initiatives, assign priority scores (RICE/MoSCoW), estimate effort, group by Now/Next/Later, and export as image or PDF. Visualization included.	Prioritization framework application (RICE · MoSCoW · ICE) · Roadmap structure · Tradeoff communication · Stakeholder presentation	React Router v6 · Dynamic route params · Chart.js or Recharts timeline · PDF/image export · TypeScript	React · React Router · Recharts or Chart.js · TypeScript · Tailwind CSS	Prioritization framework comparison doc · Roadmap spec · PRD
	■ Strong PM relevance. Use RICE on this project's own feature backlog as the PM exercise.				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Product Metrics Dashboard Prototype				
P12	Dashboard showing product KPIs: signups, activation rate, retention, churn, conversion, MRR. Mock JSON data. Multiple chart types. AARRR funnel visualization.	KPI definition · AARRR funnel design · Dashboard layout hierarchy · Metric interpretation · Data storytelling	Recharts · Custom hooks · Memoization · Skeleton loading states · TypeScript generics	React · Recharts · TypeScript · Tailwind CSS · Mock JSON	KPI definition doc · Dashboard spec · AARRR map · PRD
	■ <i>Foundation for every future analytics case study. The metrics you define here reappear throughout the journey.</i>				
	Global State Management — Shopping Cart / E-Commerce				
P13	Product listing, cart, and checkout flow. Add/remove/update quantities. Persist cart to LocalStorage. Global state via Zustand. Simulate payment confirmation UI.	Cart UX patterns · State persistence strategy · Checkout flow design · Payment failure state UX	Zustand or Redux Toolkit · Context API comparison · Selector patterns · Middleware · Optimistic UI	React · Zustand or Redux Toolkit · TypeScript · Tailwind CSS	State architecture diagram · Cart flow spec · PRD
	Next.js SSR/SSG Content Site with Headless CMS				
P14	Content-driven site (case studies, blog, or documentation) powered by a headless CMS. Static generation for performance, dynamic pages where needed, SEO optimized.	Content modeling · SEO strategy · Static vs dynamic routing tradeoffs · Non-technical stakeholder tooling design	Next.js · Static Site Generation · Server-Side Rendering · Contentful or Sanity · Image optimization · next/head SEO	Next.js · Contentful or Sanity · TypeScript · Tailwind CSS · Vercel	Content model spec · SEO checklist · Static vs dynamic decision doc · PRD
	Managed Auth & Cloud Database App (Supabase/Firebase)				
P15	Full-featured app with cloud-hosted auth (signup, login, protected routes) and real-time database via Supabase or Firebase. User-specific data isolation. Password reset flow.	Auth flow documentation · Identity lifecycle design · Security requirement writing · Non-functional requirements	Supabase or Firebase Auth · Firestore or Supabase DB · Protected routes · Real-time subscriptions · Row-level security basics	Supabase or Firebase · Next.js · TypeScript · Tailwind CSS	Auth flow diagram · Security requirements · Data model · PRD
	■ <i>Managed services first. Understand what the service gives you before building it yourself.</i>				
	Component Testing with Jest and React Testing Library				
P16	Full test suite for P9 or P13. Unit tests per component. Integration tests for key user flows. Coverage report generated and interpreted. CI test run on GitHub Actions.	Acceptance criteria as test cases · QA strategy documentation · Release confidence definition · CI gate criteria	Jest · React Testing Library · Test coverage tooling · Mocking · GitHub Actions CI pipeline intro	Jest · React Testing Library · GitHub Actions · TypeScript	Test strategy doc · Coverage report · CI pipeline spec · PRD
	■ <i>Closes Phase II. Acceptance criteria traced directly to tests. A PM who understands this writes better requirements.</i>				

Cross the full-stack line. Build custom APIs, connect to real databases, implement authentication from scratch, and understand the server-side decisions that drive product tradeoffs. Introduce file uploads, cloud storage, and transactional email as real product capabilities. PM skills introduced: API contract design, data modeling, technical requirement writing, system architecture documentation, migration planning.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Custom REST API with Node.js and Express				
P17	Standalone REST API for the P9 Task Manager. Full CRUD endpoints, request validation middleware, error handling, rate limiting, and Postman collection with documentation.	API contract documentation · Endpoint design · HTTP status code strategy · Versioning rationale · Rate limit requirement	Node.js · Express.js · RESTful routing · Middleware chain · Postman · dotenv · express-validator	Node.js · Express.js · Postman · dotenv · express-validator	API contract doc · Postman collection · Rate limit spec · PRD
	PostgreSQL Integration with Prisma ORM				
P18	Replace LocalStorage/Supabase with a custom PostgreSQL database. Schema design, migrations via Prisma, relational queries, and indexed lookups.	Data model documentation · Migration planning · Relational vs flat data tradeoffs · Schema versioning strategy	PostgreSQL · Prisma ORM · Database migrations · Schema design · SQL fundamentals · Indexed queries	PostgreSQL · Prisma · Node.js · Express.js	ERD (Entity-Relationship Diagram) · Migration log · PRD
	■ <i>SQL is a non-negotiable PM hard skill. Every data-fluent PM can write a basic SELECT query.</i>				
	User Authentication — JWT, Sessions & Password Reset				
P19	Email + password registration and login. JWT access tokens, refresh token rotation, HTTP-only cookies, protected routes, and a full password reset via email token flow.	Auth flow documentation · Session UX design · Security requirement writing · Trust and privacy spec	JWT · bcrypt · Refresh token pattern · HTTP-only cookies · Role-based route protection · SendGrid password reset	JWT · bcrypt · Node.js · Express.js · PostgreSQL · SendGrid	Auth flow diagram · Security requirements doc · PRD
	Full-Stack App — React + Node + PostgreSQL (Integrated)				
P20	First fully integrated full-stack application. React frontend consumes the custom authenticated Node/Express/PostgreSQL API. Users register, log in, and manage their own data end to end.	Full system architecture documentation · Cross-layer dependency mapping · End-to-end user journey · Deployment runbook	Full-stack integration · CORS configuration · Environment variable management · Frontend + backend co-deployment · Render or Railway hosting	React · Node.js · Express · PostgreSQL · Prisma · Render or Railway	System architecture diagram · Full-stack PRD · Deployment runbook
	■ <i>First project where every layer is yours. The debugging will teach more than any tutorial.</i>				
	Role-Based Access Control (RBAC) + Admin Dashboard				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P21	Extend P20 with admin and standard user roles. Admin dashboard with user management, content moderation, system settings, feature flag preview, and audit log of admin actions.	Permission matrix design · Admin UX patterns · Internal tooling design · Audit trail requirements · Operational visibility	RBAC middleware · Permission-gated rendering · Audit logging · Admin UI patterns · Soft delete pattern	Node.js · PostgreSQL · React · JWT with roles · Prisma	Permission matrix · Admin flow spec · Audit log spec · PRD
	File Uploads, Cloud Storage & Attachments				
P22	Users can upload files and images as task attachments. Cloud storage via AWS S3 or Supabase Storage. File type and size validation, presigned URLs, and download/preview.	File handling UX · Storage cost modeling · Upload failure state design · File lifecycle policy	AWS S3 or Supabase Storage · Multer · Presigned URLs · File validation middleware · Image optimization	AWS S3 or Supabase Storage · Multer · Node.js · React	File handling spec · Storage cost estimate · PRD
	Transactional Email System with SendGrid				
P23	Transactional emails: welcome on registration, password reset, task assignment notification, weekly digest. HTML email templates, delivery tracking, and bounce handling.	Transactional email spec · User communication design · Deliverability requirements · Email metrics (open, bounce, unsubscribe)	SendGrid API · HTML email templating · Webhook handling for delivery events · Unsubscribe management	SendGrid · Node.js · Express.js · HTML email templates · Prisma	Email flow spec · Template inventory · Delivery metrics spec · PRD
	■ <i>Every automated message is a product decision. Communication design is a PM craft.</i>				
	NoSQL Document Store — Activity Feed with MongoDB				
P24	Activity feed and event log using MongoDB. Flexible document schema, pagination, filter and sort. Structured comparison of SQL vs NoSQL tradeoffs documented as a PM artifact.	SQL vs NoSQL decision framework · Schema flexibility cost-benefit · Data architecture documentation	MongoDB · Mongoose · Document schema design · Aggregation pipeline · Indexed queries · Pagination patterns	MongoDB · Mongoose · Node.js · Express.js	SQL vs NoSQL decision doc · Schema design · Aggregation query library · PRD
	■ <i>Closes Phase III. The decision doc produced here is a real PM technical artifact.</i>				

Build the multi-user, collaborative product layer — the feature set that separates a personal tool from a team product. Teams, workspaces, invitations, real-time collaboration, comments, and a customer feedback portal. This phase introduces the complexity of shared data models, permission hierarchies, and cross-user interactions. PM skills introduced: multi-tenant scoping, collaboration UX, customer discovery systems, team onboarding flows.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Team Workspace — Multi-User Collaboration				
P25	Users create workspaces, invite team members via email, and collaborate on shared projects. Organization-level data model, member roles (owner, admin, member), and workspace switching.	Multi-user data model design · Team onboarding flow · Invitation UX · Permission hierarchy documentation	Organization/team schema · Email invitations with token validation · RBAC at org level · Workspace context middleware	Node.js · PostgreSQL · Prisma · React · SendGrid · JWT	Team data model diagram · Invitation flow spec · Permission hierarchy matrix · PRD
	Comments, Mentions & Task Threads				
P26	Threaded comments on tasks and projects. @mention notifications, emoji reactions, comment edit/delete with audit trail, and comment-level permission control.	Collaborative UX design · Notification trigger taxonomy · Mention and threading interaction spec	Nested comment schema · @mention parsing · Notification triggers · Soft delete with audit · React rich text basics	Node.js · PostgreSQL · Prisma · React · WebSockets (intro)	Comment system spec · Notification trigger map · PRD
	Real-Time Collaboration with WebSockets				
P27	Live task updates pushed to all workspace members without refresh. User presence indicators, typing status, live cursor positions on shared documents, and operational status panel.	Real-time UX spec · Presence design · Message delivery guarantee definition · Concurrent user experience mapping	WebSockets · Socket.io · Room-based event broadcasting · Presence tracking · Optimistic UI patterns	Socket.io · Node.js · PostgreSQL · React · Redis (pub/sub intro)	Real-time spec · Presence flow diagram · Latency tolerance rules · PRD
	In-App Notification Center				
P28	Multi-channel notification system: in-app notification bell with unread count, email notifications, user preference controls per notification type, and notification analytics.	Notification taxonomy design · Opt-in/opt-out UX · Notification fatigue strategy · Preference management spec	Notification schema · Unread count aggregation · WebSocket-driven badge updates · Preference storage	Node.js · PostgreSQL · Socket.io · React · SendGrid	Notification taxonomy · Preference matrix · Fatigue analysis doc · PRD
	■ <i>Notification strategy is one of the most user-trust-sensitive decisions a PM makes.</i>				
	Customer Feedback & Feature Request Portal				
P29	Public-facing portal where users submit feedback, vote on feature requests, and track status. Admins categorize, prioritize, respond, and link requests to roadmap items.	Voice-of-customer system design · Feedback synthesis methodology · Qualitative data loop · Roadmap transparency · Customer discovery documentation	Voting and ranking system · Status workflow · Public vs authenticated views · Admin moderation queue	Next.js · PostgreSQL · Prisma · React · SendGrid · Auth	Feedback taxonomy · Prioritization workflow spec · Customer discovery log · PRD

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	■ <i>Strong PM relevance. The artifacts produced here are real discovery outputs, not simulations.</i>				
	Full-Text Search with Algolia or PostgreSQL FTS				
P30	Search across tasks, projects, comments, and documents. Typo tolerance, faceted filtering, relevance ranking, search analytics (no-results rate, top queries), and zero-results handling.	Search relevance definition · Zero-results handling strategy · Search analytics interpretation · Discoverability UX	Algolia or PostgreSQL full-text search · Index schema · Relevance tuning · Faceted filtering · Search analytics	Algolia or PostgreSQL FTS · Node.js · React · TypeScript	Search spec · Relevance criteria doc · Zero-results strategy · PRD
	■ <i>Closes Phase IV. Search is a product strategy decision masquerading as a technical one.</i>				

Build products that make money and measure themselves. Stripe integration, subscription lifecycle, feature flags, behavioral analytics, A/B testing, lifecycle email, and a complete customer data platform. This phase produces the strongest PM portfolio candidates: metric optimization and GTM projects. PM skills introduced: pricing model design, funnel analysis, experiment design, KPI instrumentation, LTV/CAC modeling.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Stripe One-Time Payment Integration				
P31	One-time payment processing. Stripe Checkout or Elements. Webhook handling for payment confirmation, failure, and refund events. Purchase confirmation email. Idempotency handled.	Payment flow documentation · Checkout UX design · Payment failure state handling · PCI compliance awareness · Webhook event spec	Stripe API · Stripe Webhooks · Idempotency keys · Payment intent lifecycle · Webhook signature verification	Stripe · Node.js · Express.js · React · SendGrid	Payment flow diagram · Webhook event spec · Failure handling matrix · PRD
	Subscription Billing — SaaS Tier Model				
P32	Full subscription lifecycle: free trial, plan selection (free/pro/team), upgrade/downgrade, cancellation, reactivation. Stripe Customer Portal. Invoice history. Usage limits per plan.	Pricing tier design · Subscription lifecycle mapping · Plan entitlement logic · Churn definition · LTV/CAC fundamentals	Stripe Subscriptions · Stripe Customer Portal · Billing webhooks · Entitlement middleware · Plan upgrade/downgrade logic	Stripe · Node.js · PostgreSQL · Prisma · React	Pricing tier spec · Subscription lifecycle diagram · Entitlement matrix · PRD
	■ Subscription logic is the most common PM domain in SaaS. Own every edge case.				
	Pricing Page, Packaging & Upgrade Prompts				
	★ PORTFOLIO CANDIDATE — Cross-Functional GTM: Pricing Launch				
P33	Pricing page with plan comparison, psychological pricing design, contextual upgrade prompts within product, and billing portal link. A/B test variant of pricing copy documented.	Pricing psychology · Packaging strategy · Conversion optimization · Plan comparison UX · Upgrade prompt placement rationale	Feature-gated UI components · Contextual upgrade modals · Pricing page A/B variant structure · Stripe plan switching	React · Stripe · Tailwind CSS · TypeScript	Pricing rationale doc · Upgrade prompt spec · A/B hypothesis · PRD
	■ Portfolio candidate. Pricing decisions require engineering, marketing, sales, and legal alignment.				
	Feature Flags with LaunchDarkly				
P34	Feature flags across the product: gradual rollout, kill switch, per-user targeting, percentage rollout, and beta group targeting. Flag inventory documented and lifecycle managed.	Rollout strategy design · Flag lifecycle management · Percentage rollout risk assessment · Beta group definition	LaunchDarkly SDK · Flag targeting rules · A/B flag variants · Flag cleanup discipline · CI/CD integration	LaunchDarkly · React · Node.js · TypeScript	Flag inventory · Rollout strategy doc · Flag lifecycle policy · PRD

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Product Analytics — Event Taxonomy with Mixpanel/Amplitude				
P35	Instrument the product with behavioral event tracking. Define event taxonomy. Build activation funnel, retention cohorts, and feature adoption dashboard. Identify first drop-off point.	Event taxonomy design · Funnel definition · Activation metric selection · AARRR framework · Retention analysis	Mixpanel or Amplitude SDK · Event tracking · Funnel visualization · Cohort analysis · Segment.io intro	Mixpanel or Amplitude · Segment.io (optional) · React · Node.js	Event taxonomy doc · Funnel spec · AARRR dashboard · Activation definition · PRD
	■ <i>The event taxonomy you design here is the most important data decision a PM makes.</i>				
	Onboarding Flow Design & Activation Optimization				
	★ PORTFOLIO CANDIDATE — Metric Optimization: Activation Funnel				
P36	Multi-step onboarding wizard with progress tracking. Measure drop-off at each step. Redesign the step with the highest drop-off. Before/after activation rate comparison documented.	Onboarding funnel analysis · Activation metric definition · Drop-off hypothesis generation · Iteration decision documentation	Multi-step wizard UI · Onboarding progress persistence · Event tracking on each step · Feature flag for variant	React · Node.js · Mixpanel or Amplitude · LaunchDarkly · PostgreSQL	Onboarding funnel doc · Drop-off analysis · Redesign rationale · Before/after metrics · PRD
	■ <i>Portfolio candidate. 'How I improved activation by X%' is the clearest PM impact story.</i>				
	A/B Testing — Hypothesis to Statistical Significance				
P37	Design and run a controlled A/B test on a live feature (CTA, onboarding step, or pricing prompt). Hypothesis, sample size calculation, variant via LaunchDarkly, results, and rollout decision.	A/B test design · Statistical significance · MDE (Minimum Detectable Effect) calculation · Results interpretation · Rollout/rollback decision memo	LaunchDarkly variants · Chi-square or z-test basics · Sample size calculator · Results visualization in analytics tool	LaunchDarkly · Mixpanel or Amplitude · React · Statistics fundamentals	A/B test design doc · Results report · Statistical significance calculation · Rollout decision memo · PRD
	Lifecycle Email & Retention System				
P38	Behavioral email sequences: welcome, onboarding incomplete reminder, feature adoption nudge, weekly digest, reactivation campaign, and churn risk alert. Segmented by user behavior.	Lifecycle messaging strategy · Retention loop design · User segmentation · Email metrics (open, click, unsubscribe, conversion) · Churn prevention framework	SendGrid or Resend · Cron job scheduling · Behavioral trigger logic · BullMQ job queue intro · User segment queries	SendGrid or Resend · BullMQ · Redis · Node.js · PostgreSQL · Cron	Lifecycle email map · Segment definitions · Retention loop diagram · PRD
	SQL Analytics — Direct Database Querying for PMs				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P39	Write SQL queries against the product database to answer business questions: funnel completion rates, cohort retention, MRR by plan, churn by segment, feature adoption by role.	Data question formulation · SQL as PM self-service tool · Analyst collaboration simulation · KPI storytelling from raw data	Advanced SQL · Window functions · CTEs · Aggregation queries · Data visualization in Metabase or Redash	PostgreSQL · Metabase or Redash · SQL · pgAdmin	SQL query library · KPI dashboard · Data question log · PRD
	■ A PM who can write SQL does not wait three days for an analyst to answer a question.				
	Customer Data Platform — Segment.io + Data Export				
P40	Integrate Segment.io as a central event router connecting analytics, email, and CRM. Add CSV and PDF data export for users and admin reporting. Single event fires to all downstream tools.	CDP architecture · Data governance basics · Single source of truth design · Export UX · Reporting dashboard design	Segment.io · Source and destination config · Identity resolution · Event routing · CSV/PDF export libraries	Segment.io · Node.js · React · Mixpanel · SendGrid · pdf-lib or similar	CDP architecture diagram · Event routing map · Export spec · PRD
	■ Closes Phase V. Every growth-stage company uses a CDP. Knowing how it works makes you a better PM.				

Phase VI · P41
– P47

REAL-TIME, SCALE & ADVANCED INFRASTRUCTURE

Build products that operate at concurrency and scale. Background processing, caching, performance optimization, integrations marketplace, accessibility, and internationalization. This phase produces enterprise-grade technical fluency and a second GTM portfolio candidate. PM skills introduced: SLA writing, latency tolerance, infrastructure cost-benefit, i18n product strategy, integration marketplace design, and WCAG compliance documentation.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Background Job Queue with BullMQ and Redis				
P41	Async background job processing: email delivery, report generation, image processing, and data exports. Job retry logic, dead letter queue, and admin job status dashboard.	Async vs sync decision framework · Job failure UX · SLA definition for background tasks · Operational status design	BullMQ · Redis · Worker processes · Retry/backoff patterns · Dead letter queue · Job status WebSocket updates	BullMQ · Redis · Node.js · PostgreSQL · React	Job taxonomy · Failure handling spec · SLA document · PRD
	Redis Caching & Performance Optimization				
P42	Add Redis caching to high-read endpoints. Cache invalidation strategy, TTL design, cache-hit rate measurement, before/after latency comparison, and Google Lighthouse performance audit.	Performance requirement writing · Cache strategy documentation · Latency SLA definition · Page speed to conversion rate connection	Redis caching · TTL strategy · Cache invalidation patterns · Google Lighthouse audit · Web Vitals baseline	Redis · Node.js · Google Lighthouse · Postman · pgAdmin	Performance baseline report · Cache strategy doc · Web Vitals dashboard · PRD
	Integrations Marketplace — OAuth2 & Webhooks				
	★ PORTFOLIO CANDIDATE — GTM: Integrations Launch				
P43	Integration hub connecting to external services (Slack, Google Calendar, GitHub, or Zapier-style webhooks). OAuth2 flows for each integration, outbound webhooks, and an integration management UI.	Integration product design · OAuth2 UX · Webhook reliability spec · Partner integration documentation · API ecosystem strategy	OAuth2 authorization flow · Webhook payload handling · Integration state management · Retry logic for outbound webhooks	OAuth2 · Node.js · PostgreSQL · React · Slack API or Google API	Integration catalog · OAuth2 flow diagrams · Webhook spec · PRD
	■ <i>Integrations are a platform strategy decision. This project introduces ecosystem thinking.</i>				
	Internal Sales Enablement Portal				
P44	Internal-facing portal with product positioning guides, objection handling scripts, competitive comparisons, launch FAQs, customer segment profiles, and demo scripts. Searchable and role-gated.	GTM planning · Sales enablement documentation · Stakeholder alignment artifacts · Influence without authority · Launch readiness checklist	Next.js · Admin roles · Markdown/CMS content · Search · Authentication · PDF export	Next.js · PostgreSQL · Prisma · Auth · Markdown · Search	GTM kit · Sales enablement guide · Competitive comparison matrix · Launch checklist · PRD
	■ <i>The artifacts here are real GTM outputs. A hiring manager reviewing this understands cross-functional execution.</i>				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Accessibility Audit & Remediation (WCAG 2.2 AA)				
P45	Full WCAG 2.2 Level AA audit of the existing product. Automated audit via axe-core, manual keyboard navigation testing, screen reader testing, color contrast fixes, focus management, and ARIA implementation.	Accessibility as product requirement · WCAG 2.2 compliance documentation · Audit to remediation prioritization · Inclusive design principles	axe-core · ARIA · Keyboard navigation · Focus management · Screen reader testing (NVDA or VoiceOver) · Color contrast tooling	axe-core · NVDA or VoiceOver · Colour Contrast Analyser · React · Node.js	WCAG 2.2 audit report · Remediation priority matrix · Compliance sign-off checklist · PRD
	■ WCAG 2.2 AA is non-negotiable on all PMD projects. This project builds the audit skill formally.				
	Internationalization & Localization (i18n/l10n)				
P46	Multi-language support: string externalization, locale switching, RTL layout support (Arabic/Hebrew), date/number/currency formatting per locale, and a translation workflow with Crowdin or similar.	Localization requirement writing · Market entry documentation · Translation workflow design · RTL product considerations · Regional pricing strategy	i18next · React-i18next · RTL CSS · Locale-aware formatting · Pluralization rules · Translation file management	i18next · React · Node.js · Crowdin (optional) · Tailwind CSS RTL	Localization spec · Locale support matrix · Translation workflow · Market entry doc · PRD
	Progressive Web App (PWA) — Mobile Bridge				
P47	Convert the existing product to a PWA: service worker, offline mode, install prompt, push notifications, and app-shell architecture. Bridge between web and native mobile.	Offline-first UX requirements · PWA vs native decision framework · Push notification strategy · App install conversion design	Service worker · Cache API · Web Push API · Manifest.json · Lighthouse PWA audit · Background sync	Service Worker API · Web Push API · Workbox · React · Lighthouse	Offline UX spec · PWA vs native decision doc · Push notification taxonomy · PRD
	■ Closes Phase VI. PWA is the most cost-effective mobile strategy for a solo product builder.				

Integrate AI as a deliberate product capability — not as a novelty, but as a solution to a defined user problem. LLM integration, structured outputs, vector search, RAG pipelines, document analysis, and agentic workflows. This phase produces the strongest 0-to-1 greenfield portfolio candidates. PM skills introduced: AI product scoping, prompt engineering as product design, hallucination risk documentation, AI cost modeling, responsible AI requirement writing, and agentic workflow design.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	OpenAI API Integration — Text Generation Feature				
P48	AI-powered text generation in an existing product: task description generation, summary creation, or meeting notes. Streaming responses, token tracking, rate limiting, and cost monitoring.	AI feature scoping · Prompt as product design · Cost-per-feature analysis · AI failure mode documentation · Human-in-the-loop design	OpenAI API · Streaming responses · Token budgeting · Rate limiting · Error classification · Cost monitoring	OpenAI API · Node.js · React · TypeScript	AI feature spec · Prompt library v1 · Cost model · Failure mode doc · PRD
	Anthropic Claude API — Structured Output Product				
P49	Product feature using Claude producing structured JSON for downstream consumption. Multi-turn conversation state, system prompt engineering, output schema validation, and evaluation rubric.	Structured output spec writing · System prompt as product requirement · AI output evaluation criteria · Tradeoff: open-ended vs structured responses	Anthropic SDK · System prompts · Multi-turn conversation management · JSON schema · Zod validation · Output testing	Anthropic SDK · Node.js · React · TypeScript · Zod	System prompt spec · Output schema · Evaluation rubric · PRD
	AI PRD Assistant — Greenfield 0-to-1 Product				
	★ PORTFOLIO CANDIDATE — 0-to-1 Greenfield: AI PRD Assistant				
P50	AI assistant that turns vague product ideas into structured PRDs: problem statement, personas, user stories, success metrics, risks, and open questions. Human-in-the-loop editing. Versioned outputs.	Full PM lifecycle execution · Product-market fit validation · AI UX design · Prompt engineering as product requirement · 0-to-1 product launch	OpenAI or Anthropic API · Streaming · Structured output parsing · Version history in PostgreSQL · React rich text editing	Anthropic or OpenAI API · Next.js · PostgreSQL · Prisma · Stripe · TypeScript	Market research · Personas · Full PRD · MVP prioritization (RICE) · Launch plan · Case study
	■ <i>Strongest greenfield portfolio candidate. Solves a problem hiring managers recognize immediately.</i>				
	Vector Database & Semantic Search				
P51	Embed a document corpus and build semantic search over it. Similarity scoring, hybrid search (keyword + semantic), relevance evaluation, and a comparison of embedding models.	Semantic vs keyword search tradeoffs · Relevance evaluation design · Embedding cost analysis · Search quality metrics	Pinecone or pgvector · OpenAI Embeddings API · Cosine similarity · Hybrid search · Chunking strategy	Pinecone or pgvector · OpenAI Embeddings · Node.js · TypeScript	Search architecture diagram · Relevance evaluation doc · Embedding cost model · PRD

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	RAG Pipeline — Retrieval-Augmented Generation				
P52	RAG system over a document corpus: ingest, chunk, embed, retrieve, and generate. Document Q&A; with source citations, hallucination mitigation strategy, and context window management.	RAG architecture documentation · Hallucination risk spec · Citation UX design · Grounding strategy · Context length tradeoffs	LangChain or custom RAG · Vector retrieval · Context window management · Source citation injection · Chunking experiments	LangChain · Pinecone or pgvector · OpenAI or Anthropic · Node.js · TypeScript	RAG architecture diagram · Hallucination mitigation spec · Chunking strategy doc · PRD
	AI Document Analysis Tool				
P53	Upload a PDF or contract. AI extracts structured data (key clauses, dates, parties), generates summaries, answers questions, and flags risks. Multi-document comparison mode.	Document processing UX · Extraction accuracy definition · Edge case documentation · Confidence scoring design	PDF text extraction · Multimodal inputs · Structured extraction prompts · Confidence scoring · pypdf or pdfplumber	Anthropic or OpenAI API · pypdf · Node.js · React · TypeScript	Extraction spec · Accuracy evaluation rubric · Edge case log · PRD
	AI Chatbot with Memory, Persona & Guardrails				
P54	Persistent AI assistant with conversation memory, defined persona, tool use (web search or calculator), content guardrails, and a user feedback loop for quality improvement.	Persona specification · Guardrail requirement writing · Feedback loop design · Trust and safety spec · Content policy documentation	Conversation memory patterns · Tool use / function calling · Content filtering · Feedback storage · Conversation history in PostgreSQL	Anthropic or OpenAI API · LangChain or custom · PostgreSQL · React · TypeScript	Persona spec · Guardrail spec · Content policy doc · Feedback loop design · PRD
	ML Predictive Feature — Churn Prediction or Recommendation				
P55	Train and deploy a simple ML model as a product feature. Churn prediction or content recommendation. Feature engineering, model evaluation, A/B test against rule-based baseline, and model card.	ML feature requirements · Model evaluation criteria · False positive/negative cost analysis · Explainability requirement · Model card documentation	Python · scikit-learn · Feature engineering · Model evaluation metrics (AUC, F1) · FastAPI model serving	Python · scikit-learn · pandas · FastAPI · PostgreSQL	ML feature spec · Model card · A/B test design · Performance metrics report · PRD
	■ <i>First Python project. PM must understand model tradeoffs to write good ML requirements.</i>				
	AI Agent with Tool Orchestration				
P56	Autonomous agent that plans, uses tools (search, code execution, API calls), and completes multi-step tasks with minimal human input. Full observability, tracing, and failure recovery built in.	Agentic product design · Human-in-the-loop spec · Observability requirements · Failure recovery design · Agentic UX patterns	LangChain agents or custom · Tool definitions · Reasoning loop patterns · LangSmith or similar tracing · Error recovery	Anthropic or OpenAI API · LangChain · Python or Node.js · LangSmith	Agent architecture diagram · Tool inventory · Observability spec · Failure recovery spec · PRD
	■ <i>Closes Phase VII. Agentic AI is the current frontier. This project is deliberately placed after deep foundations.</i>				

Phase VIII · P57
– P66+

PLATFORM, MOBILE & PRODUCTION SCALE

Graduate from product builder to platform thinker. GraphQL, React Native, Docker, CI/CD pipelines, micro-frontend architecture, multi-tenant SaaS, observability, and a full SaaS launch simulation. Portfolio milestone projects emerge from this phase. The capstone project is the culminating full-stack SaaS product that becomes the flagship portfolio showcase. PM skills introduced: platform strategy, mobile PM, API versioning, infrastructure cost modeling, enterprise documentation, DevOps collaboration, and end-to-end launch narrative.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	GraphQL API with Apollo Server				
P57	Replace or augment REST endpoints with GraphQL. Schema design, resolvers, mutations, real-time subscriptions, and N+1 optimization via DataLoader. Apollo Client integration.	GraphQL vs REST decision documentation · Schema design as product contract · API versioning strategy · Query efficiency tradeoffs	GraphQL · Apollo Server · Resolvers · DataLoader · Schema-first design · Apollo Client · TypeScript	GraphQL · Apollo Server · Apollo Client · Node.js · TypeScript	API schema doc · REST vs GraphQL decision memo · Schema governance policy · PRD
	React Native Mobile App (Cross-Platform iOS & Android)				
P58	Port core product features to iOS and Android via React Native. Navigation, device APIs (camera, location), push notifications (APNs/FCM), offline mode, and app store submission prep.	Mobile-specific UX requirements · Platform parity decisions · App store submission spec · Offline-first design · Mobile-web feature gap analysis	React Native · Expo · React Navigation · Device APIs · APNs/FCM push · TypeScript · Offline storage	React Native · Expo · TypeScript · React Navigation · Firebase Cloud Messaging	Mobile PRD · Platform parity matrix · App store submission checklist · Push notification spec · PRD
	Containerization with Docker & Docker Compose				
P59	Containerize the full-stack application. Multi-container setup with Docker Compose (app, PostgreSQL, Redis, worker). Production-ready Dockerfile patterns with multi-stage builds.	Infrastructure requirement writing · Environment parity documentation · Deployment risk log · Onboarding reduction via containerization	Docker · Docker Compose · Multi-stage builds · Environment variable management · Health checks · Volume management	Docker · Docker Compose · Node.js · PostgreSQL · Redis	Deployment architecture diagram · Environment spec · Docker runbook · PRD
	CI/CD Pipeline with GitHub Actions				
P60	Full CI/CD pipeline: automated tests on every PR, build validation, staging deploy, manual production gate with approval, rollback procedure, and zero-downtime deploy confirmed.	Release process documentation · Deploy gate criteria · Rollback decision tree · Change management spec · DORA metrics awareness	GitHub Actions · Workflow YAML · Test automation in CI · Environment secrets management · Blue/green deployment concept	GitHub Actions · Docker · Render or Railway or AWS	CI/CD pipeline diagram · Deploy runbook · Rollback procedure · Gate criteria doc · PRD
	Multi-Tenant SaaS Architecture				
P61	Extend the product to support multiple organizations (tenants). Row-level security for data isolation, tenant-scoped API, org-level admin, per-tenant billing via Stripe, and tenant onboarding flow.	Multi-tenant scoping · Data isolation requirement writing · Tenant onboarding flow · Per-tenant pricing model · Enterprise sales readiness	PostgreSQL row-level security · Tenant context middleware · Schema-per-tenant vs shared schema decision · Stripe per-tenant billing	PostgreSQL · Node.js · Prisma · Stripe · React	Tenant architecture diagram · Data isolation spec · Onboarding flow · Pricing per tenant · PRD

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	■ <i>Multi-tenancy is the architectural foundation of every B2B SaaS product.</i>				
	Micro-Frontend Architecture				
P62	Decompose the frontend into independently deployable micro-frontends using Webpack Module Federation. Shared design system, independent deployment pipelines, and cross-team ownership simulation.	Platform architecture documentation · Team topology design · API boundary definition · Cross-team dependency charting · Feature ownership model	Webpack Module Federation · Shared component library · Independent deployment pipelines · Shell application pattern	Webpack Module Federation · React · TypeScript · GitHub Actions	Platform architecture diagram · Team boundary spec · Feature ownership matrix · PRD
	Observability Stack — Logging, Metrics, Alerts & Incident Response				
P63	Instrument the production application with structured logging, error tracking, performance metrics, uptime monitoring, on-call alerting, and a formal incident response runbook.	SLA and SLO definition · Incident response documentation · On-call runbook writing · DORA metrics (MTTR, deployment frequency) · Post-mortem process	Structured logging · Sentry error tracking · Uptime monitoring · Alert threshold design · PagerDuty or similar	Sentry · Datadog or Grafana · PagerDuty or similar · Node.js · GitHub Actions	SLA/SLO doc · Alert threshold spec · Incident runbook · Post-mortem template · PRD
	Advanced Reporting & Executive Dashboard				
	★ PORTFOLIO CANDIDATE — Analytics PM: Executive Dashboard				
P64	Executive-level reporting dashboard: activation, retention, churn, MRR, feature adoption, feedback volume, NPS simulation, A/B experiment results, and SQL-driven custom reports. CSV export.	Executive reporting design · Business intelligence fundamentals · Cohort analysis · KPI storytelling · Metrics hierarchy (leading vs lagging)	Advanced SQL · Metabase or Redash · Custom charting · Cohort analysis queries · CSV/PDF export · Data modeling	PostgreSQL · Metabase or Redash · Recharts · Node.js · React · SQL	Metrics hierarchy doc · Dashboard spec · SQL query library · Cohort analysis report · PRD
	Enterprise Usage Tracking & Consumption Pricing Migration				
	★ PORTFOLIO CANDIDATE — Cross-Functional GTM: Pricing Migration				
P65	Migrate from flat subscription to consumption-based pricing. Usage metering, real-time consumption telemetry, usage alerts, overage billing, enterprise pricing negotiation tooling, and GTM launch kit.	GTM launch kit construction · Internal alignment documentation (Engineering/Sales/Legal/Support) · Pricing migration risk log · Post-launch feedback loop · Change management	Usage metering · Consumption telemetry logging · Stripe usage-based billing · Backward compatibility strategy · Real-time usage dashboard	Stripe Usage-Based Billing · PostgreSQL · Node.js · React · Segment.io	GTM launch kit · Pricing migration spec · Internal FAQ · Sales enablement update · Risk log · PRD
	Full SaaS Product Launch Simulation — Capstone				
	★ FLAGSHIP PORTFOLIO PROJECT — Full SaaS Capstone				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P66	Combine the strongest features into a cohesive, production-grade SaaS product: landing page, auth, onboarding, AI assistant, roadmap builder, feedback portal, subscription billing, analytics, admin, CI/CD, and complete GTM materials. The flagship portfolio showcase.	End-to-end product lifecycle ownership · Full case study suite · Launch narrative · Cross-functional artifact production · Business model documentation	Full stack synthesis · All prior capabilities integrated · Production deployment · Monitoring live · Performance budget met · WCAG 2.2 AA confirmed	Next.js · TypeScript · PostgreSQL · Prisma · Stripe · OpenAI/Anthropic · Mixpanel · Segment · Resend · Vercel · GitHub Actions · Docker · Figma	Complete PMD artifact suite · Case study · Portfolio showcase entry · Launch retro
	■ This threshold is earned, not scheduled. The coach assesses readiness from the quality of P55–P65 retrospectives.				

RECOMMENDED READING & OFFLINE ASSIGNMENTS — BY PHASE

Phase I–II · Foundation & Front-End	• Don't Make Me Think — Steve Krug · UX fundamentals every PM must internalize before building anything • The Pragmatic Programmer — Hunt & Thomas · Developer mindset and craft habits for non-developers • HTML & CSS: Design and Build Websites — Jon Duckett · Visual learner's illustrated entry point to the web • CSS: The Definitive Guide — Meyer & Weyl · Reference-grade CSS depth for when layouts break
Phase II–III · React & Backend	• Clean Code — Robert C. Martin · Writing code your future self and future teammates will not curse • The Pragmatic Programmer — Hunt & Thomas (revisit) · Deeper on systems thinking after first backend • Learning React — Alex Banks & Eve Porcello · React mental model from first principles • Designing Data-Intensive Applications — Martin Kleppmann · The PM's database and distributed systems bible
Phase IV–V · Analytics & Growth	• Lean Analytics — Croll & Yoskovitz · AARRR, the one metric that matters, and cohort thinking • Hacking Growth — Ellis & Brown · Systematic growth experimentation from the people who invented it • Trustworthy Online Controlled Experiments — Kohavi, Tang, Xu · A/B testing done rigorously at Microsoft and Google • The Mom Test — Rob Fitzpatrick · How to talk to users without getting lied to
Phase IV–V · Product Strategy	• Inspired — Marty Cagan · How the best product teams in the world actually work • Continuous Discovery Habits — Teresa Torres · Opportunity trees, assumption testing, and user interview discipline • Competing Against Luck — Clayton Christensen · Jobs-to-be-Done at full depth and nuance • Obviously Awesome — April Dunford · Positioning as a PM discipline — what category are you actually in?
Phase VI–VII · AI, Platform & Scale	• Building Machine Learning Powered Applications — Emmanuel Ameisen · ML products from prototype to production • AI Product Management — A Practical Guide · Scoping, ethics, and failure modes of AI features • Platform Revolution — Parker, Van Alstyne, Choudary · Network effects, platform strategy, and ecosystem design • The Phoenix Project — Kim, Behr, Spafford · DevOps and engineering culture through a business novel
Phase VII–VIII · Leadership & Business	• Measure What Matters — John Doerr · OKRs in practice at Google and beyond • The Lean Startup — Eric Ries · Build-measure-learn for every stage and every scale • Crossing the Chasm — Geoffrey Moore · Technology adoption lifecycle and the B2B PM challenge • High Output Management — Andrew Grove · Management fundamentals for PMs operating with engineers • Thinking in Bets — Annie Duke · Decision-making under uncertainty — the PM's core operating condition
Offline Practice — Throughout the Journey	• PM Exercises (pmexercises.com) · Daily product design and metric practice problems • Lenny's Newsletter (lennyrachitsky.com) · Best practitioner-written PM content available today • Stratechery (stratechery.com) · Ben Thompson's business and technology strategy analysis — read every piece • Product teardown habit: pick one app per month, write a 500-word friction audit · Builds pattern recognition faster than any course • User interview practice: conduct 2 real user interviews per project using The Mom Test methodology • SQL practice: LeetCode Database or Mode Analytics SQL Tutorial · 30 minutes per day during Phases III–V

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